

Zheng XU

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Education

University of Manchester

Sep. 2023 – Present

BEng (Hons) Electrical and Electronic Engineering

- Results: First Class (Expected)
- Core Modules: Control Systems I & II, Digital Signal Processing, Microcontroller Engineering I & II, Computer Systems Architecture, Concurrent Systems, VLSI & High Speed Digital Design, Digital Systems Design I & II, Electronic Circuit Design I & II, Sensors & Instrumentation, C Programming, Numerical Analysis

Research Projects

BCI-based Robotic Arm Control System (Deep Reinforcement Learning)

Sep. 2025 – Present

University of Manchester Final Year Project

- Designed EEGTransformer, a hybrid CNN–Transformer architecture for motor imagery EEG classification, achieving 73.80% (BCI IV-2a, 4-class), 82.87% (IV-2b, 2-class), and 88.78% (PhysioNet, 109 subjects) accuracy across three benchmark datasets.
- Implemented a two-stage transfer learning protocol (cross-subject pre-training + subject-specific fine-tuning) that improved classification accuracy by 32% with minimal calibration data (~90 trials per user).
- Conducted systematic channel reduction study from 64 to 8 electrodes for OpenBCI deployment; identified C3 as the most critical motor cortex electrode and retained 72.54% accuracy.
- Validated preprocessing pipeline through controlled ablation: proved 8–30 Hz bandpass filtering improves accuracy by +18.44%, justifying exclusion of ICA.
- Engineered a Transformer-based Deep Q-Network (SequenceTransformerDQN) for continuous control, mapping noisy 4-class MI signals to 8-direction planar trajectories; achieved a 99% simulated reach rate through robust closed-loop error compensation.
- Developed a complete Sim2Real pipeline by constructing a unified OpenAI Gym-compatible interface, bridging PyBullet simulation and the physical SO-101 robotic arm to enable seamless zero-shot deployment of RL policies.
- Implemented real-time serial control algorithms for the multi-joint manipulator, featuring synchronized joint velocity profiling, dynamic trajectory interpolation, and soft-limit boundary protections to ensure precise and smooth hardware actuation.

Hai'an Institute of Intelligent Equipment, Shanghai Jiao Tong University

Jun. 2025 – Jul. 2025

Research Intern

- Conducted research on robotic arm control systems, focusing on motion accuracy and dynamic response optimisation.
- Designed and implemented control algorithms in MATLAB and embedded C++ for trajectory planning and feedback regulation.
- Prepared experimental reports and presented findings to the laboratory's research supervisors.

Embedded Systems Project

Sep. 2024 – Apr. 2025

- Designed and implemented an autonomous line-following robot using an STM32 Nucleo-F401RE microcontroller.
- Integrated multiple sensors (TCRT5000 IR, MPU-6050 gyroscope, HC-SR04 ultrasonic) for real-time environmental perception.
- Developed a PID control algorithm in C++ for accurate path-tracking and obstacle avoidance.
- Designed custom PCBs in Altium Designer; performed sensor calibration and signal conditioning to reduce noise.

- Implemented Bluetooth Low Energy (BLE) communication for wireless data logging and remote debugging.

Quanser Aero 2 Control Systems Project

Jun. 2023 – Jul. 2023

- Modelled and analysed the Quanser Aero 2 dual-rotor platform using MATLAB and Simulink.
- Identified rotor and pitch subsystem dynamics; estimated system parameters (gain, damping ratio, natural frequency).
- Designed and tuned P, PI, and PID controllers; compared simulated and experimental responses.
- Demonstrated proficiency in system identification, control theory, and real-time hardware-in-the-loop testing using QUARC.

Skills & Certifications

- **Programming:** Python (PyTorch, MNE-Python, NumPy), C/C++, MATLAB, VHDL, Assembly
- **Embedded Systems:** STM32, PIC series, real-time control, sensor integration, PCB design (Altium Designer)
- **Simulation & Modelling:** MATLAB/Simulink, LabVIEW, SolidWorks, PyBullet, Sentaurus TCAD
- **Machine Learning:** Deep reinforcement learning (DQN), Transformer architectures, transfer learning, EEG signal processing
- **Languages:** Chinese (native), English (proficient, TOEFL 106/120)

Certifications:

- Certified MATLAB Associate – MathWorks